

EEE (15) 6035:
REVISION:2015

Reg No _____
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**SIXTH SEMESTER DIPLOMA EXAMINATION IN ELECTRICAL AND
ELECTRONICS ENGINEERING
ELECTRICAL MACHINE DESIGN**

Maximum marks :100

Time: 3 Hours

PART – A
(Max. Marks: 10)

Marks

I. Answer all questions in one or two sentences. Each question carries 2 marks.

1. Define rating of an electric machine.
2. Write any two essential differences between magnetic and electric circuits.
3. Define transformation ratio.
4. Write the output equation and output co-efficient of synchronous machine.
5. Name the material used for brush in a dc machine.

(5x2=20)

PART B

(Max. Marks: 30)

II. Answer any five of the following questions. Each question carries 6 marks.

1. Write any two essential differences between magnetic and electric circuits.
2. Give classification of insulating materials.
3. Explain the use of CRGOS strip wound transformer core.
4. List the special features of the cage rotor of an induction machine?
5. List the advantages of revolving field system in synchronous machine.
6. Give the importance of compensating winding in dc machine.
7. Mention the different modes of operation of a dc Machine.

(5x6=30)

PART C

(Max. Marks: 60)

(Answer one full question from each unit. Each full question carries 15 marks)

UNIT I

- III (a) Explain about the classification of magnetic materials. (8)
(b) Explain about the factors which limit the design of an electric machine. (7)

Or

- IV. (a) Explain the major considerations in Electrical Machine Design (8)
(b) List the properties of a good insulating material. (7)

UNIT II

- V. (a) Derive output equation for 1 phase transformer. (8)
(b) Compute main dimensions D and L of a 3.7 KW, 400V, 3-phase, 4-pole, 50 Hz Squirrel cage induction motor. $B_{av} = 0.45 \text{ Wb/m}^2$ Electrical loading = 23000 Amp-conductors/meter, Efficiency = 85%, Power factor = 0.84, Winding factor = 0.955, Stacking factor = 0.9. (7)

Or

- VI. (a) List the special features of the cage rotor an induction machine? (8)
(b) Estimate the main dimensions including winding conductor area of a 3-phase, delta to star core type transformer rated at 300KVA, 6600/440 V, 50 Hz. A suitable core with three steps having a circumscribing circle of 0.25 m diameter and a leg spacing of 0.4 m is available. $\delta = 2.5 \text{ A/mm}^2$, E.M.F. per turn = 8.5 V, $K_w = 0.28$, $S_f = 0.9$ (space factor). (7)

UNIT III

- VII. (a) Explain the choice of specific magnetic and electric loadings of synchronous machines. (8)
(b) Explain types of synchronous machines (7)

Or

- VIII. (a) Explain how does damper winding improve the performance of a synchronous machine (8)
(b) A 2 pole, 50 Hz turbo alternator has a core length of 1.5 m. the mean flux density over the pole pitch is 0.5 Wb/m^2 , the stator ampere conductors per cm are 260 and peripheral speed 100 metre/second. The average span of the stator coils is one pole pitch. Determine the output which can be obtained from the machine. (7)

UNIT IV

IX.(a) Derive output equation of a dc machine (8)

(b) A 4 pole, 25 HP, 500 V, 600 rpm series motor has an efficiency of 82%. The pole faces are square and the ratio of pole arc to pole pitch is 0.67. Take $B_{av} = 0.55 \text{ Wb/m}^2$ and $A_c = 17000 \text{ amp.cond./m}$. Obtain the main dimensions of the core and particulars of a suitable armature winding. (7)

Or

X.(a) Explain how to select the number of poles of a dc machine (8)

(b) The commutator of a 10 pole, 1000 KW, 500 Volt, 300 rpm D.C. generator has 450 segments and an external diameter of 1 meter. Determine a suitable axial length for the commutator, give details of brushes with regard to commutation and temperature rise. Assume current density as 6 amps/cm^2 , voltage drop due to brush contact as 2.2 volt, brush pressure as 1250 Kg/m^2 and co-efficient of friction as 0.25 (7)